$$(2x+3y+4z)(4x^2+9y^2+16z^2-6xy-12yz-8zx)$$
$$=(2x)^3+(3y)^3+(4z)^3-3(2x)(3y)(4z)$$
$$=8x^3+27y^3+64z^3-72xyz$$

Example 3.23 Evaluate $10^3-15^3+5^3$

Solution

We know that, if $a+b+c=0$, then $a^3+b^3+c^3=3abc$

Here, $a+b+c=10-15+5=0$

Replace a by 10, b by –15, c by 5

Therefore, $10^3+(-15)^3+5^3=3(10)(-15)(5)$

$$10^3-15^3+5^3=-2250$$

Exercise 3.4

1. Expand the following:
   (i) $(x+2y+3z)^2$ (ii) $(-p+2q+3r)^2$
   (iii) $(2p+3)(2p-4)(2p-5)$ (iv) $(3a+1)(3a-2)(3a+4)$
2. Using algebraic identity, find the coefficients of x^2, x and constant term without actual expansion.
   (i) $(x+5)(x+6)(x+7)$ (ii) $(2x+3)(2x-5)(2x-6)$
3. If $(x+a)(x+b)(x+c)=x^3+14x^2+59x+70$, find the value of
   (i) $a+b+c$ (ii) $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}$ (iii) $a^2+b^2+c^2$ (iv) $\frac{a}{bc}+\frac{b}{ac}+\frac{c}{ab}$
4. Expand:
   (i) $(3a-4b)^3$ (ii) $\left(x+\frac{1}{y}\right)^3$
5. Evaluate the following by using identities:
   (i) 98^3 (ii) 1001^3
6. If $(x+y+z)=9$ and $(xy+yz+zx)=26$, then find the value of $x^2+y^2+z^2$.
7. Find $27a^3+64b^3$, if $3a+4b=10$ and $ab=2$.
8. Find x^3-y^3, if $x-y=5$ and $xy=14$.
9. If $a+\frac{1}{a}=6$, then find the value of $a^3+\frac{1}{a^3}$.
10. If $x^2+\frac{1}{x^2}=23$, then find the value of $x+\frac{1}{x}$ *and* $x^3+\frac{1}{x^3}$.
11. If $\left(y-\frac{1}{y}\right)^3=27$, then find the value of $y^3-\frac{1}{y^3}$.

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